PROJECT REPORT

TITLE:

MOOD-BASED ACTIVITY RECOMMENDATION SYSTEM (CLI)

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INTRODUCTION:-

This project is a CLI-based application developed using Python. It helps users by suggesting activities based on their current mood. The system interacts with the user, takes input, and provides suitable recommendations.

MOTIVATION OF THE PROJECT :-

Students often feel confused about what to do when they are stressed, bored, or feeling low. This project was created to help users make better decisions based on their emotional state.

PROBLEM STATEMENT:-

To design a system that takes user mood as input and suggests appropriate activities to improve productivity and mental well-being.

OBJECTIVE OF THE PROJECT:-

1) To build a simple AI-based recommendation system.
2) To understand user input handling.
3) To provide useful activity suggestions.
4) To implement basic decision-making logic.

EXISTING METHODS:-

Currently, people manually decide what to do based on mood, or use random apps without personalized suggestions.

PROS AND CONS OF EXISTING METHODS:-

Pros:

- Easy to use.
- No technical knowledge required.

Cons:

- No personalization.
- Not always effective.
- Time-consuming decision making.

HARDWARE AND SOFTWARE REQUIREMENTS:-

Hardware:

- Laptop / PC.

Software:

- Python.
- Command Prompt / Terminal.
- GitHub.

METHODOLOGY AND GOAL:-

The system uses a rule-based approach where user input is taken and processed using conditional statements. Based on the input, the system suggests relevant activities.

FUNCTIONAL MODULES DESIGN AND ANALYSIS:-

- Input Module (user enters mood).
- Processing Module (decision-making using logic).
- Output Module (displaying recommendations).

- Validation Module (handling incorrect input).

ALGORITHM DEVELOPMENT:-

1) Start the program.
2) Take user name.
3) Display mood options.
4) Take user input.
5) Validate input (max 3 attempts).
6) Suggest activities based on mood.
7) Ask for feedback.
8) Repeat or exit.

CODING:-

```
name = input("Enter your name: ")

while True:
    # Displaying mood options
    print("Hii,", name, "\nHow are you feeling today?")
    print("1. Happy / Energetic")
    print("2. Sad / Low")
    print("3. Stressed / Anxious")
    print("4. Bored")

    attempts = 0

    # Loop for input validation (max 3 attempts)
    while True:
        if attempts >= 3:
            print("\nToo many invalid attempts. Exiting program.")
```

```
        exit()

    user_input = input("Enter your choice (1-4): ")

    # Checking if input is a number
    if user_input.isdigit():
        choice = int(user_input)

        # Checking if number is in valid range
        if choice in [1, 2, 3, 4]:
            break
        else:
            print("Invalid choice, please try again.")
    else:
        print("Please enter a valid number.")

    attempts += 1


print(f"\nHey {name}, here are your recommended activities:\n")

# Suggesting activities based on mood
if choice == 1:
    print("• Do workout or sports")
    print("• Start a new project")
    print("• Hang out with friends")

elif choice == 2:
    print("• Listen to music")
    print("• Take 15-minute walk")
    print("• Call a trusted friends or family members")
```

```
    elif choice == 3:
        print("• Practice deep breathing")
        print("• Try meditation")
        print("• Take a break and relax")

    elif choice == 4:
        print("• Try a new hobby")
        print("• Solve puzzles or games")
        print("• Explore something new")


    print("\nTip: Your mood affects your productivity, so choose
wisely.")

    # Feedback loop (validates yes/no input)
    while True:
        feedback = input("\nWas this suggestion helpful? (yes/no):
").lower()

        if feedback == "yes":
            print("Great! I'm glad it works for you")
            break
        elif feedback == "no":
            print("Thanks for your feedback, we will improve!")
            break
        else:
            print("Invalid input, please enter 'yes' or 'no'.")

    # Asking user if they want to continue
    while True:
        again = input("\nDo you want to check another mood?
(yes/no): ").lower()
```

```
if again == "yes":
    break
elif again == "no":
    print("\nThank you for using the system!")
    exit()
else:
    print("Invalid input, please enter 'yes' or 'no'.")
```

OUTPUT:-

1.Program Start Screen:

```
==== RESTART: C:/Users/SAKSHAM JAIN/AppData/Local/Programs/Python/Python313/mood.py ====
Enter your name: Saksham
Hii, Saksham
How are you feeling today?
1. Happy / Energetic
2. Sad / Low
3. Stressed / Anxious
4. Bored
Enter your choice (1-4):
```

2.Invalid Input Handling:

```
Enter your name: Saksham
Hii, Saksham
How are you feeling today?
1. Happy / Energetic
2. Sad / Low
3. Stressed / Anxious
4. Bored
Enter your choice (1-4): abcd
Please enter a valid number.
Enter your choice (1-4):
```

3.Attempt Limit:

```
Hii, Saksham
How are you feeling today?
1. Happy / Energetic
2. Sad / Low
3. Stressed / Anxious
4. Bored
Enter your choice (1-4): 7
Invalid choice, please try again.
Enter your choice (1-4): 8
Invalid choice, please try again.
Enter your choice (1-4): 9
Invalid choice, please try again.

Too many invalid attempts. Exiting program.
```

4.Happy Mood Choice:

```
Enter your choice (1-4): 1

Hey Saksham, here are your recommended activities:

• Do workout or sports
• Start a new project
• Hang out with friends

Tip: Your mood affects your productivity, so choose wisely.

Was this suggestion helpful? (yes/no): yes
Great! I'm glad it works for you
```

5.Sad Mood Choice:

```
Do you want to check another mood? (yes/no): yes
Hii, Saksham
How are you feeling today?
1. Happy / Energetic
2. Sad / Low
3. Stressed / Anxious
4. Bored
Enter your choice (1-4): 2

Hey Saksham, here are your recommended activities:

• Listen to music
• Take 15-minute walk
• Call a trusted friends or family members

Tip: Your mood affects your productivity, so choose wisely.

Was this suggestion helpful? (yes/no):
```

6.Stressed Mood Choice:

```
Do you want to check another mood? (yes/no): yes
Hii, Saksham
How are you feeling today?
1. Happy / Energetic
2. Sad / Low
3. Stressed / Anxious
4. Bored
Enter your choice (1-4): 3

Hey Saksham, here are your recommended activities:

• Practice deep breathing
• Try meditation
• Take a break and relax

Tip: Your mood affects your productivity, so choose wisely.

Was this suggestion helpful? (yes/no): yes
Great! I'm glad it works for you
```

KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM:-

- Mood-based recommendation.
- Input validation system.
- Attempt limit feature.
- Feedback system.
- Loop functionality.

SIGNIFICANT PROJECT OUTCOMES:-

- Improved decision-making.
- Better user interaction.
- Practical understanding of AI basics.

TESTING AND REFINEMENT:-

The system was tested with different inputs including valid and invalid cases. Errors were handled using validation logic and attempt limits.

PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS:-

This system can be used in mental health apps, productivity tools, and student assistance systems.

CONTRIBUTION / FINDINGS OF THE PROJECT:-

- Learned basic AI concepts.
- Improved coding skills.
- Understood user interaction.

LIMITATION / CONSTRAINTS OF THE SYSTEM:-

- Limited mood options.
- No machine learning used.
- Static recommendations.

CONCLUSION:-

This project successfully demonstrates a simple AI-based system that helps users by suggesting activities based on mood.

FUTURE ENHANCEMENTS:-

- Add more moods.
- Use machine learning.
- Store user history.
- Add emoji - based mood selection

REFERENCES:-

- Python Documentation.
- Online tutorials.
- Class notes.